Polymeric Chemistry
Rheology

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Contents

1 Theory

In rheology, the flow behaviour of a substance under different forces is studied. This behaviour can be determined using rotational or oscillatory measurements.

1.1 Basic concepts

To measure rheological properties, the sample is placed between two plates, the upper one is capable of moving. When this plate is set in motion, the sample experiences a force and moves along with it, but exhibits a velocity gradient from the lower to the upper plate. A visual representation of this is shown in Figure 1.

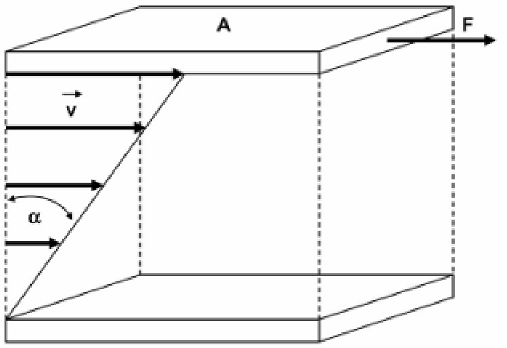

Figure 1: The diagram shows two plates with an area A, the upper one of which exerts a force F on the specimen, causing it to experience a velocity gradient $\vec{v}$ and to move through an angle α. [1]

Key parameters that can be detected or specified in this context include deformation γ, deformation rate $\dot{\gamma}$, shear stress τ and steady-state shear viscosity η. These parameters are defined by Equations (1–4).

$$\text{Deformation:} \quad \gamma \equiv \frac{dx}{dy} = \tan\alpha \tag{1}$$

$$\text{Deformation rate:} \quad \dot{\gamma} \equiv \frac{d\gamma}{dt} \tag{2}$$

$$\text{Shear stress:} \quad \tau = \frac{F}{A} = \eta \cdot \dot{\gamma} \tag{3}$$

$$\text{Steady-state shear viscosity:} \quad \eta = \frac{\tau}{\dot{\gamma}} \tag{4}$$

x and y represent the coordinate axes, with x pointing in the direction of the force F. [1]
If a strain rate is specified and the shear stress is thereby detected, the viscosity can also be determined, allowing these parameters to be investigated as a function of the strain rate. Three types of curve behaviour can be detected for the shear stress or viscosity. The parameters may follow a linear trend or deviate from it. If the parameters deviate, they may become either larger or smaller at higher deformation rates. If a sample exhibits a linear relationship, it is referred to as a Newtonian fluid. If the parameters increase at higher deformation rates, it is referred to as a non-Newtonian shear-thickening fluid; conversely, if they decrease, it is referred to as a non-Newtonian shear-thinning fluid.

The following graphs visualize the effects described above for both viscosity and shear stress.

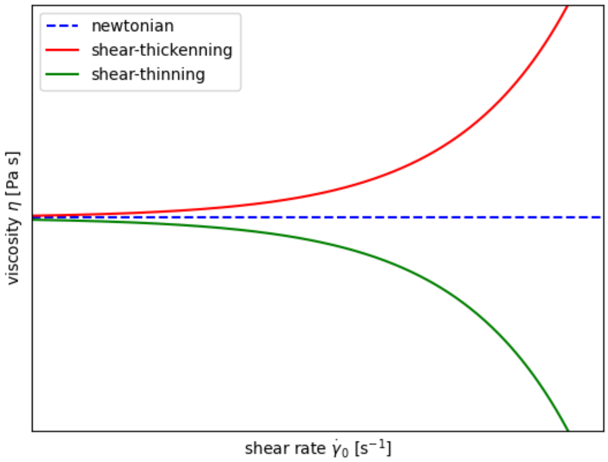

Figure 2: The picture shows the shear thinning, shear thickening and newtonian effect of the shear rate on the viscosity.

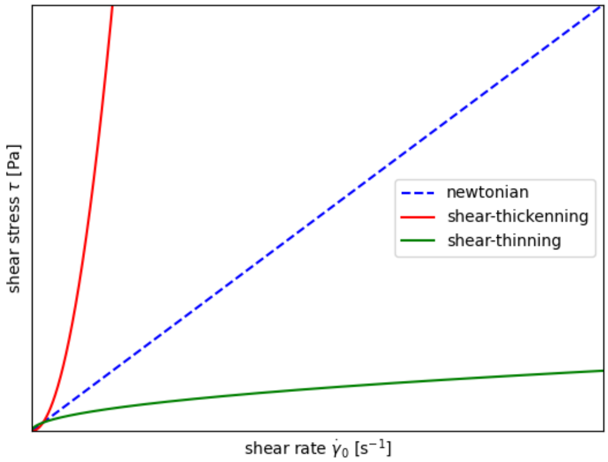

Figure 3: The picture shows the shear thinning, shear thickening and newtonian effect of the shear rate on the shear stress.

1.2 Behaviour of matter

1.2.1 Elastic

The elastic behaviour of a material can be compared to that of a rubber band or a spring. When a force is applied, causing deformation or shear stress, the material responds by undergoing shear or deformation, and returns to its original state once the force is removed. [1]

1.2.2 Viscous

In the case of viscous behaviour, the material deforms continuously when a force is applied and retains its deformation once the force is removed. [1]

1.2.3 Plastic

In the case of plastic behaviour, the material only deforms once a certain stress level is reached; prior to this, it behaves either elastically or shows no deformation. [1]

1.2.4 Viscoelastic

Most materials exhibit viscoelastic behaviour, which describes a combination of elastic and viscous behaviour. When a stress is applied, the material will deform elastically and then continue to deform under further shear stress. If this stress is removed, the deformation is partially reversed, but a slight deformation remains. This residual deformation depends on whether the material exhibits more elastic or viscous behaviour.

1.2.5 Thixotropy and Rheopexy

If the rate of deformation is specified and the viscosity is measured over time, the material may behave differently. In this case, the material may exhibit time-dependent viscosity. If the viscosity of the material decreases under load, it takes time to return to its initial viscosity once the load is removed, this is referred to as thixotropy. If the initial viscosity is never reached, this is referred to as non-thixotropic behaviour. The viscosity may also increase under load, in which case the term rheopexy is generally used.

1.3 Measurement methods

As mentioned earlier, in rheological measurements the sample is placed between two plates and moved through the upper plate. However, in a plate-plate geometry, the rate of deformation of the material depends on the radius. For this reason, the upper plate is often replaced by a slightly conical plate. This conical plate can then be rotated or oscillated. During rotation, the conical plate can be rotated at different speeds, with either the strain rate or the stress being applied. This allows the viscosity, strain rate and shear stress to be determined. During oscillation, the stress or strain is again applied, and the other quantity (strain or stress) is thereby determined. Furthermore, the cone plate is oscillated only up to a certain deflection and at a specific frequency. This is used to measure the complex viscosity η, the viscoelasticity and, at the same time, the extent to which the material is more viscous or elastic. The entire process is determined by varying the deflection and subsequently selecting the frequency, and follows the equations below.

1.4 Complex parameters of oscillation

When determining the behaviour of the material, a stress or a strain is first specified. In oscillation measurements, the complex strain is given as

$$\gamma^* = \gamma_0 \cdot \exp(\mathrm{i} \cdot \omega \cdot t). \tag{5}$$

A similar calculation is used for the complex shear stress, given as

$$\tau^* = \tau_0 \cdot \exp(\mathrm{i} \cdot \omega \cdot t + \delta). \tag{6}$$

Here, γ_0 and τ_0 describe the amplitude of the oscillation for the strain and stress, ω the frequency, and δ describes the phase shift between the two quantities in the time domain and, at certain values, characterizes the material. If δ takes the value 0, the material behaves elastically; a value of $\frac{\pi}{2}$ corresponds to viscous behaviour, and anything in between corresponds to viscoelastic behaviour.

The quotient of shear stress and strain describes the complex modulus, which is expressed as a complex number

$$G^* = \frac{\tau^*}{\gamma^*} = G^{'} + \mathrm{i} \cdot G^{''} \tag{7}$$

The quantity $G^{'}$ describes the storage modulus and is defined as

$$G^{'} = \frac{\tau_0}{\gamma_0} \cdot \cos(\delta). \tag{8}$$

$G^{''}$, on the other hand, is the loss modulus and is given as

$$G^{''} = \frac{\tau_0}{\gamma_0} \cdot \sin(\delta). \tag{9}$$

The quotient of these values corresponds to the tangens of the phase shift δ.

$$\tan\delta \equiv \frac{G^{''}}{G'} \tag{10}$$

If the calculation yields a number less than 1, the elastic behaviour dominates; conversely, the viscous behaviour dominates.

In addition, the complex viscosity, which is given as

$$|\eta^*| = \frac{\sqrt{G'^2 + G''^2}}{\omega} = \frac{\tau_0}{\gamma_0 \cdot \omega}, \tag{11}$$

is measured. [1]

2 Experimental part

The initial measurements involved rotational measurements. First, the flow behavior of honey, a 14was investigated. For the investigation, a varying shear rate of $\dot{\gamma} = 0,1$ to $500\,\mathrm{s}^{-1}$ was applied to the honey, a shear rate of $\dot{\gamma} = 5000$ to $1\,\mathrm{s}^{-1}$ to the PVP solution, and a shear rate of $\dot{\gamma} = 1$ to $100\,\mathrm{s}^{-1}$ to the starch-water suspension.

3 Results

3.1 Rotational measurements

After analyzing the different graphs it can be stated that honey has the typical linear progression which speaks for a newtonian fluid. By analyzing Figure 4 in the area from 10^{-1}- 10^{0} of the deformation velocity a non linear part can be observed. However, this does not mean that honey is a non-Newtonian fluid, but rather that it takes a certain amount of time before measurement accuracy can be guaranteed. This comes from the first "loosening" of the molecules, because these can be intertwinded or stuck in each other. This can also be applied on the other measurements.

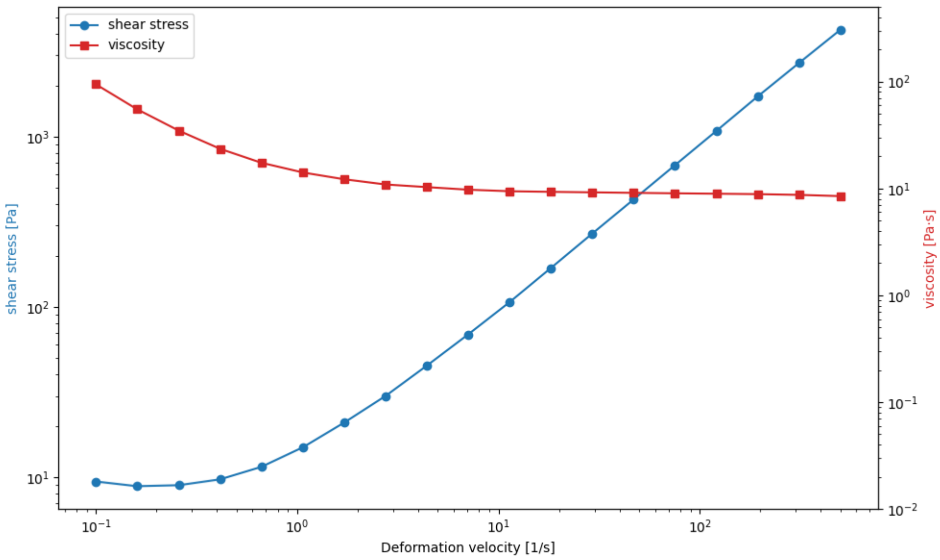

Figure 4: The graph shows the measurements of honey. On the left y-axis the shear stress and on the right y-axis the viscosity is plotted, both of these are plotted against the deformation velocity.

By applying the same procedure from the honey a clear trend for the 14%-PVP solution can be observed. Figure 5 shows a tiny decline in the end which implies a shear thinning non newtonian fluid. Meaning it has a viscosity and shear stress decrease with increasing deformation velocity. This is also confirmed after research [2].

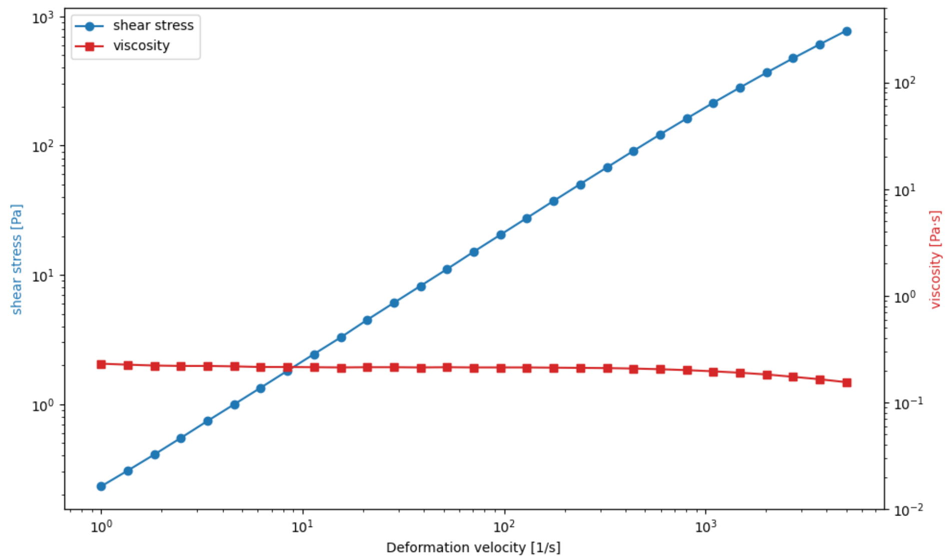

Figure 5: The graph shows the measurements of 14%-PVP solution. On the left y-axis the shear stress and on the right y-axis the viscosity is plotted, both of these are plotted against the deformation velocity.

The next measurement of this exercise was the measurement of a 50% starch solution. Figure 6 shows a clear exponential function which points to a non newtonian shear thickening fluid. Meaning the viscosity and shear stress is increasing when increasing the deformation velocity. This is not that surprising because starch is also the main ingredient of oobleck and the properties are pretty known.

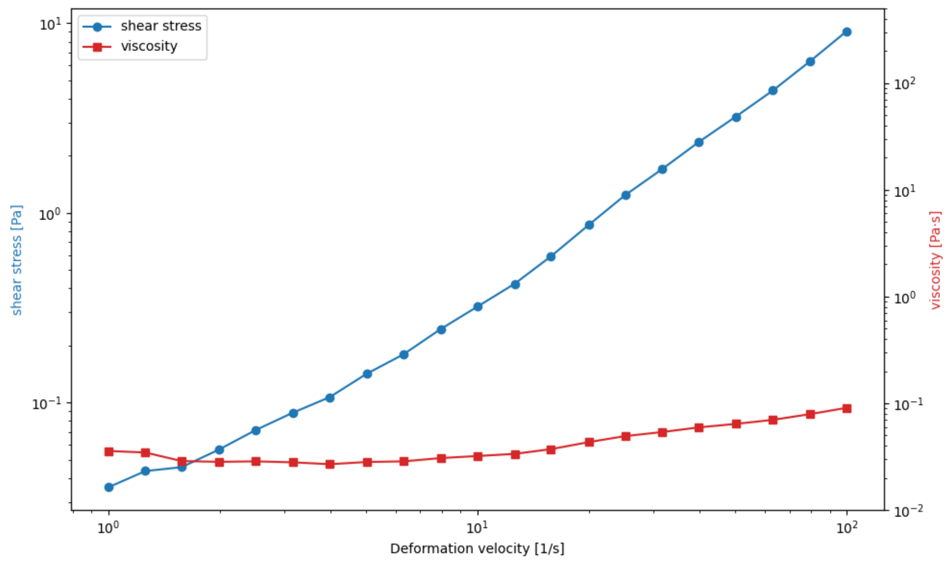

Figure 6: The graph shows the measurements of a starch solution. On the left y-axis the shear stress and on the right y-axis the viscosity is plotted, both of these are plotted against the deformation velocity.

3.2 Time dependant measurement of the viscosity

The next task was to determine the viscosity of ketchup in the dependency over time. This measurement was repeated for three times under the same conditions. After examination of Figure 7, a difference between the viscosity before and after applying the high deformation velocity is visible. This behaviour is typicall for thixotropic liquids. Because these have shear thinning properties after deformation velocity is applied.

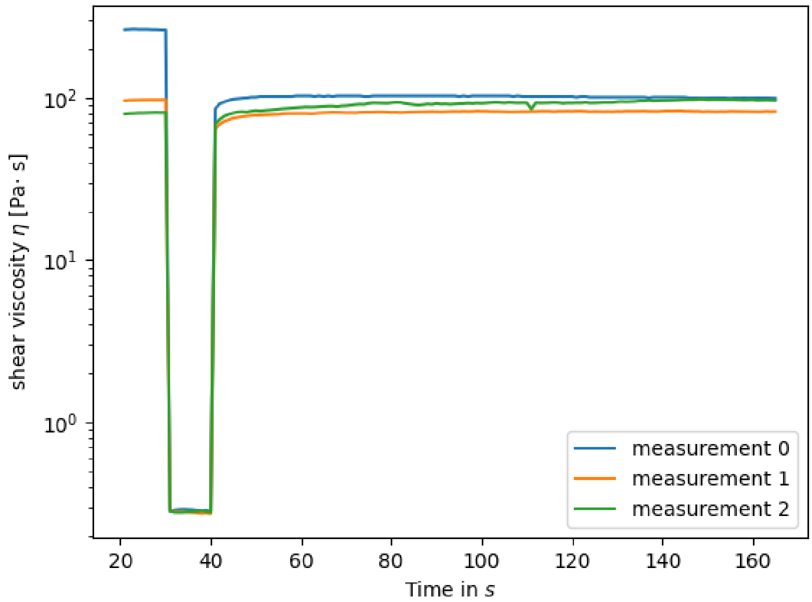

Figure 7: The graph shows the three time dependant viscosity measurements of ketchup. The shear viscosity is plotted against the duration.

3.3 Oszillational measurements to determine the viscoelastic properties

This experiment was split into two different parts one Amplitude sweep where the frequency was constant and one frequency sweep where the amplitude was constant.

3.3.1 Strain sweep

After conducting the experiment where 20%-PVP solution was used, here the storage and the loss modules where observed and measured and were plotted against the amplitude [9],[10]. Also the complex viscosity was observed and plotted against the amplitude [8]. By observing the Graphs [9] and [10] the linear viscoelastic range can be observed at the parts of the graph where the curves stay constant. After observation a deformation of 72% could be concluded and an amplitude of 70% was used in the next experiment. This procedure can be explained with the analogy to DSC and TGA where TGA is done first to get info on the highest temperature before the polymer degrades and this tempereature is used for the DSC. In this case the Amplitude sweep is done to determine the highest amplitude possible, which then can be used in the frequency sweep.

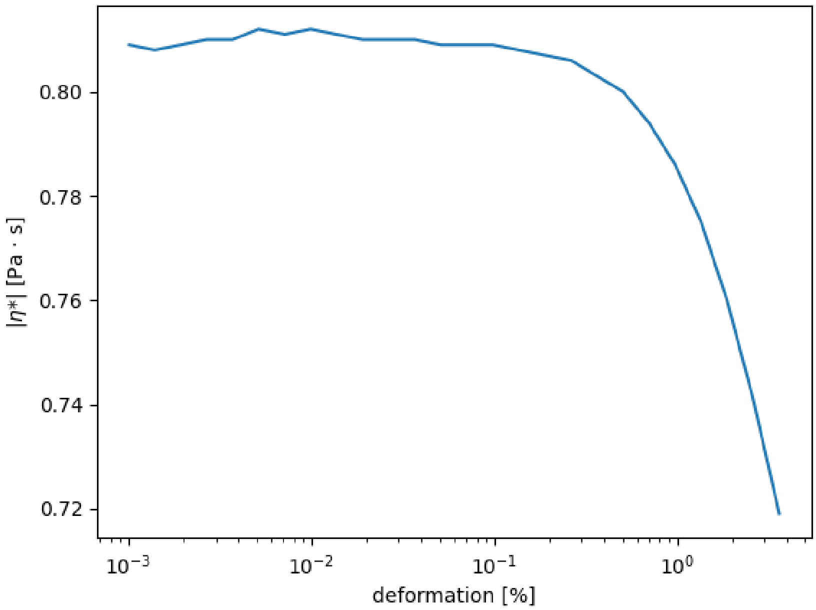

Figure 8: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant frequency.

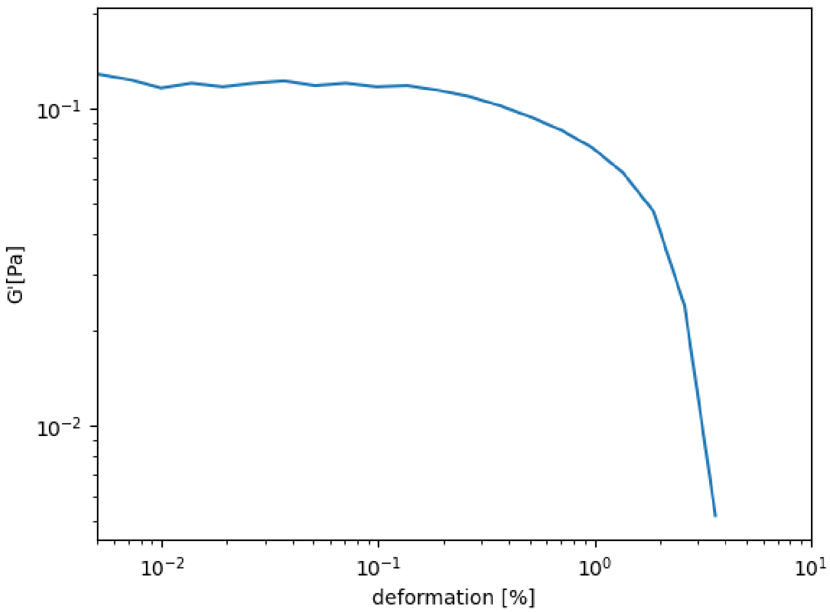

Figure 9: This graph shows the storage module G' plotted against the deformation of 20%-PVP solution at constant frequency.

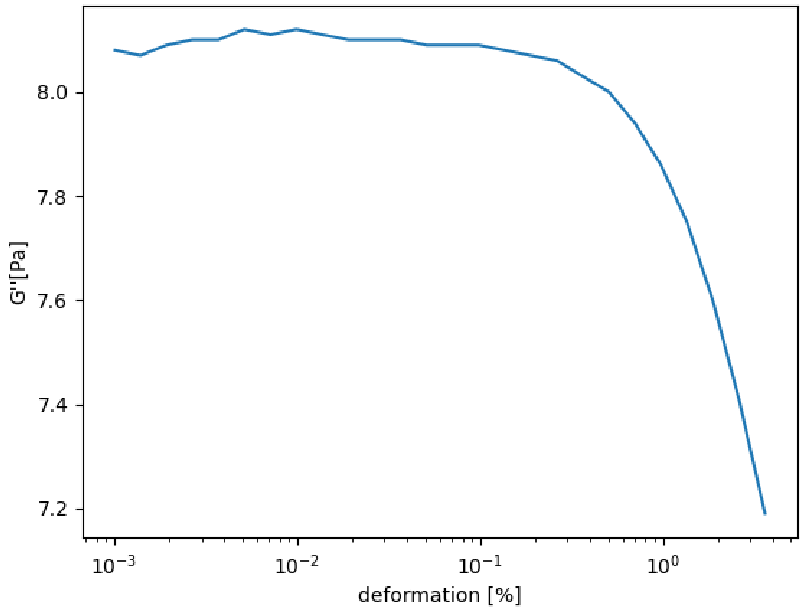

Figure 10: This graph shows the loss module G" plotted against the deformation of 20%-PVP solution at constant frequency.

3.3.2 Frequency sweep

Like before briefly said in this part the amplitude was constant and the frequency was changed. Same as in the stress sweep the storage and the loss module where observed [12]. And the value of $\eta*$ could be determined [11]. In this case when the frequency is increased the storage and the loss moduel also increase, by closer examination a stronger increase can be observed at the storage curve. This results in a decreasing loss factor, which also results that the elastic properties get more present. In diffrent to the beginning (low frequency), where the viscous properties are more present.

These observations conclude in the fact, that with increasing frequency the chains of the PVP get more organized. What leads to a decrease in the loss factor which also leads to a decrease of the viscosity [11].

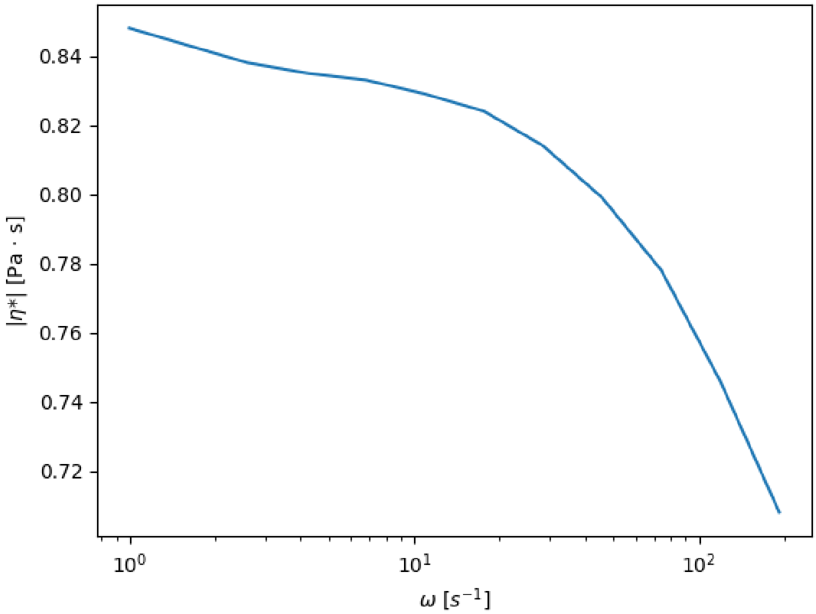

Figure 11: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant amplitude.

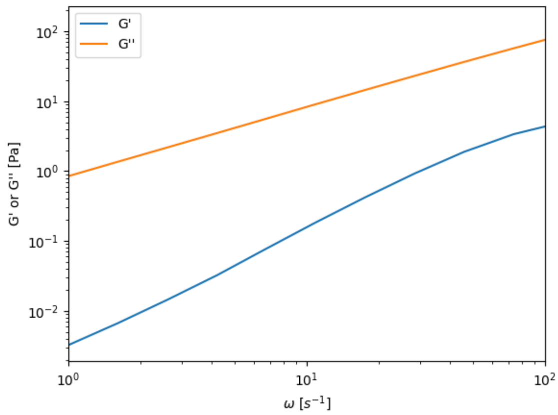

Figure 12: This graph shows the storage module G' loss module G" plotted against the deformation of 20%-PVP solution at constant amplitude.

4 Questions

4.1 Question 1

Another possible way to determine the viscosity is the viscosimetry in methode the viscosity of the measuered fluid is analyzed. This is done by the time a fluid needs to travle a certain distence in consideration to gravity.

4.2 Question 2

The flow and viscosity diagram for different fluids are shown in Figures 2 and 3.

4.3 Question 3

The biggest difference are the shear rates wich differ between those two methods. That said the shear rate of the Cone-plate is constant at all points. This is because of the radius it has to turn. The plane-plane on the other hand doesn't have that, ant therefore is better vor bigger volumina. It also is more robust if the measured substance has particles or if the substance is a gel or paste.

References

[1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.

[2] J. E. Martín-Alfonso, E. Číková, M. Omastová, *Composites Part B: Engineering* **2019**, *169*, 79–87.